

LLMs Can Play (Global) Games

Natural-language coordination, validation, and extensions

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The story in five steps

1. Start with a **canonical global game**.
2. Embed the private signal in **natural language**.
3. Show LLM actions reproduce the **Morris–Shin threshold pattern**.
4. Validate the signal channel: **scramble collapses, flip reverses**.
5. Then use the laboratory for extensions: communication and surveillance.

A coup game: coordination creates multiplicity

Following the standard coup-game framing: citizens want to participate only if the uprising succeeds; success requires participation above a threshold.

	Uprising succeeds	Uprising fails
Join	B	$-C$
Stay	0	0

Strategic complementarity

Joining is attractive when I expect many others to join.

Selection problem

Because each person's best response depends on expected participation, both mass joining and mass abstention can be self-confirming.

Enter global games

Global games make the payoff-relevant fundamental uncertain and privately observed.

- ▶ Nature draws θ , the regime's strength.
- ▶ Lower θ : easier to topple. Higher θ : harder to topple.
- ▶ Citizen i sees a noisy private signal:

$$x_i = \theta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- ▶ Agents infer both the fundamental and what others are likely to have seen.

Key move: noisy private signals generate a unique cutoff equilibrium: join when the signal says the regime is weak enough.

Canonical continuum version

Each citizen in a continuum chooses $a_i \in \{0, 1\}$, where $a_i = 1$ means JOIN; θ is the normalized participation threshold.

Aggregate participation and success

$$A \equiv \int_0^1 a_i di \quad \text{and} \quad \text{uprising succeeds} \iff A > \theta.$$

Morris–Shin's result is a unique monotone equilibrium:

Threshold rule

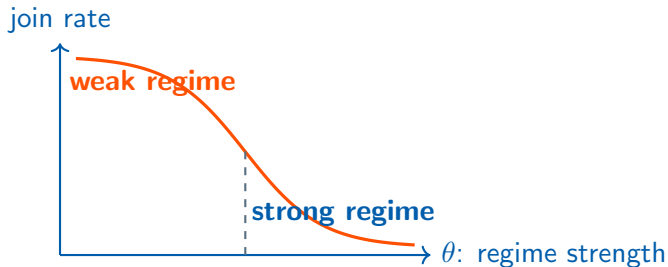
$$\text{join} \iff x_i < x^*$$

- ▶ Lower signal x_i means a weaker-looking regime.
- ▶ The cutoff x^* is common in equilibrium and pinned down by marginal-agent indifference.
- ▶ Aggregate joining therefore falls as regime strength rises.

Prediction for the LLM experiment

If citizens follow the cutoff rule, aggregate joining is pinned down by the distribution of private signals:

$$A(\theta) = \Pr(x_i < x^* \mid \theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right).$$



Empirical test: as θ rises, fewer signals fall below x^* , so JOIN rates should decline like $\Phi((x^* - \theta)/\sigma)$.

Why LLMs make this a clean laboratory

Human subjects

- ▶ costly at scale
- ▶ have memory across rounds
- ▶ hard to rewrite information environment cleanly
- ▶ difficult to expose to many counterfactual regimes

LLM calls

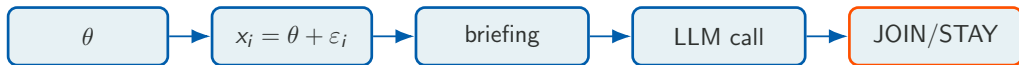
- ▶ stateless calls: no memory unless we give it
- ▶ natural language information, not just numbers
- ▶ exogenous prompt manipulation
- ▶ identical game under censorship/surveillance/placebo variants

Interpretation

The claim is not that LLMs are citizens; it is that they let us test controlled comparative statics in a rich language environment.

Related: Horton (2023), "Large Language Models as Simulated Economic Agents," NBER WP 31122.

The LLM experiment



The continuous signal becomes a private natural-language intelligence briefing.

Example: signal embedded in prose

PRIVATE BRIEFING

BOTTOM LINE: *The regime's cohesion is visibly fraying.*

ATMOSPHERE: *Street talk has shifted from complaint to anticipation.*

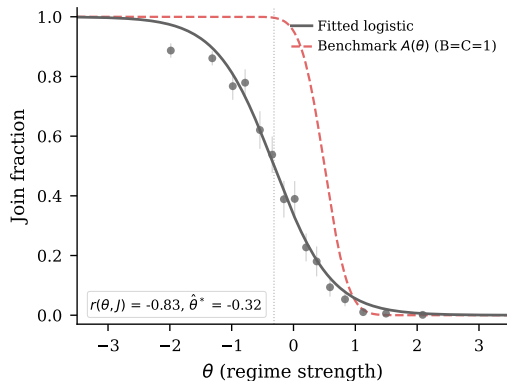
EVIDENCE:

- *Security forces show friction and delays.*
- *Business families are moving capital with urgency.*
- *Graffiti appeared overnight and has not been cleaned.*

Takeaway

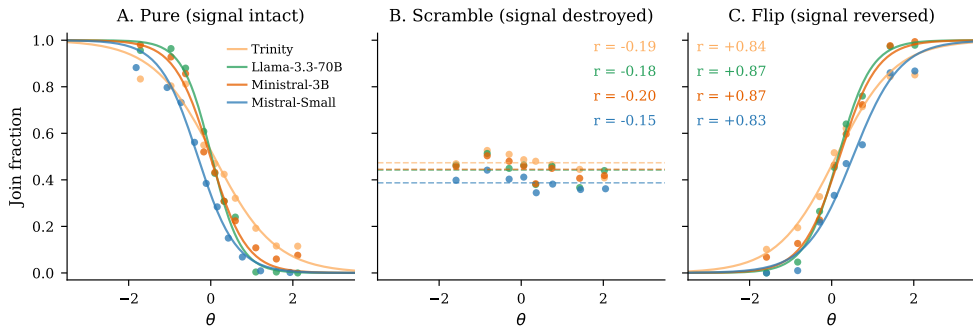
No payoff numbers. No explicit θ . The strategic signal is carried by language.

Baseline result: LLMs play the canonical global game



Join rates move with the latent fundamental and follow the predicted decreasing sigmoid.

Validation: not just generic sentiment



- ▶ **Pure:** threshold relationship appears.
- ▶ **Scramble:** destroy signal assignment, relationship collapses.
- ▶ **Flip:** reverse signal content, relationship flips.

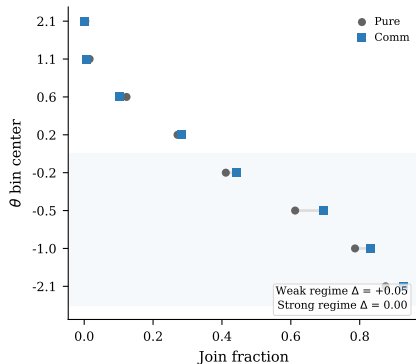
Extension 1: allow citizens to communicate

Canonical model: **signals** $\rightarrow$ **actions**.

Extension: **signals** $\rightarrow$ **peer messages** $\rightarrow$ **actions**.

- ▶ Agents send short messages to neighbours before choosing JOIN/STAY.
- ▶ Decision-stage agent sees its own briefing plus neighbours' messages.
- ▶ The communication channel can reveal what others know.

Extension 1 result: communication raises participation



Participation rises when agents can talk.

Why communication should matter

In a global game, my private signal is useful.

But your message helps me infer **what others are likely seeing and doing**.

Interpretation

Communication partly converts private information into shared information, making coordinated action easier.

Extension 2: monitor the communication channel

Add one sentence to the **messaging prompt**:

Surveillance warning

“You have reason to believe that your communications are being monitored by regime security services, and messages deemed subversive could have serious consequences for you and your contacts.”

The decision prompt is unchanged.

The agent deciding JOIN/STAY has no direct memory of the surveillance warning.

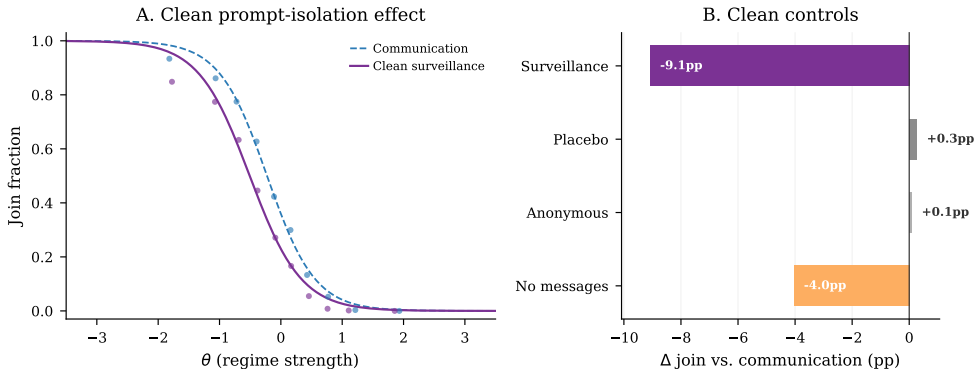
Stress test: isolate the message channel



Key design feature

Each LLM call is stateless. If actions change, the treatment must be operating through changed message content.

Extension 2 result: monitoring lowers participation



Participation falls when messages are monitored.

Mechanism clue: beliefs do not move much

The surprising part is not just lower action.

It is that stated beliefs about others' joining move much less than actions.

	Communication	Surveillance
Participation	higher	lower
Beliefs about others	similar	similar

What “beliefs” means here

After choosing JOIN or STAY, each agent forecasts the crowd:

Second-order belief

“What fraction of the other citizens do you think will join the uprising?”

- ▶ A stated expectation about **others’ participation**.
- ▶ Not a direct readout of private type or true inner state.
- ▶ Measured after action, so I treat it as descriptive rather than a clean mediator.

Key comparison

Do agents **act differently** while reporting similar beliefs?

Mechanism: monitored speech becomes less informative

Monitoring does not need to convince agents that the uprising is hopeless.

It can work by making pro-uprising agents **speak less clearly**.

Communication poisoning

The regime degrades the peer-information channel, so agents coordinate less even when their stated beliefs barely move.

What monitoring does to speech

Private signal

“The regime’s cohesion is visibly fraying.”

“Street talk has shifted from complaint to anticipation.”

Message when monitored

“Hard to know what will happen.”

“Best to stay cautious for now.”

Monitored speech becomes less informative.

The extension turns a validation result into a stress test for coordination through language.

What the evidence shows

1. **Baseline:** LLMs play the canonical global game in natural language.
2. **Validation:** actions move with latent fundamentals, not generic sentiment.
3. **Extension 1:** communication increases coordination.
4. **Extension 2:** monitoring communication lowers participation.
5. **Mechanism clue:** monitored speech changes more than stated beliefs.

The general claim: LLMs can be used as controlled agents in language-rich global games.

Discussion.

Appendix

Solving the canonical global game

Appendix: complete information gives multiplicity

Normalize the attack payoff as

$$u(1) = \begin{cases} 1 - c, & \text{if collapse occurs,} \\ -c, & \text{otherwise,} \end{cases} \quad u(0) = 0, \quad 0 < c < 1.$$

If everyone observes θ :

- ▶ If everyone attacks, $A = 1$ and collapse occurs for $\theta < 1$.
- ▶ If nobody attacks, $A = 0$ and collapse fails for $\theta > 0$.

Multiplicity

For $0 < \theta < 1$, both $A = 1$ and $A = 0$ are self-confirming equilibria.

Appendix: cutoff strategy and aggregate attacks

Add private noisy signals:

$$x_i = \theta + \sigma \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

Guess a monotone cutoff strategy:

$$a_i(x_i) = \begin{cases} 1, & x_i \leq x^*, \\ 0, & x_i > x^*. \end{cases}$$

Then, conditional on true θ ,

$$\begin{aligned} A(\theta) &= \Pr(x_i \leq x^* \mid \theta) \\ &= \Pr\left(\varepsilon_i \leq \frac{x^* - \theta}{\sigma}\right) \\ &= \Phi\left(\frac{x^* - \theta}{\sigma}\right). \end{aligned}$$

Appendix: boundary and marginal indifference

Collapse occurs when $A(\theta) \geq \theta$. The boundary fundamental θ^* solves

$$\Phi\left(\frac{x^* - \theta^*}{\sigma}\right) = \theta^*.$$

At the cutoff, the marginal agent is indifferent:

$$\Pr(\text{collapse} \mid x_i = x^*) = c.$$

Since collapse occurs for $\theta \leq \theta^*$,

$$\Pr(\theta \leq \theta^* \mid x_i = x^*) = c.$$

With a diffuse prior and normal noise,

$$\theta \mid x_i \sim \mathcal{N}(x_i, \sigma^2), \quad \Phi\left(\frac{\theta^* - x^*}{\sigma}\right) = c.$$

Appendix: clean solution

From marginal indifference,

$$\Phi\left(\frac{\theta^* - x^*}{\sigma}\right) = c \quad \Rightarrow \quad x^* = \theta^* - \sigma\Phi^{-1}(c).$$

Plug into the boundary equation:

$$\begin{aligned}\theta^* &= \Phi\left(\frac{x^* - \theta^*}{\sigma}\right) \\ &= \Phi(-\Phi^{-1}(c)) = 1 - c.\end{aligned}$$

Therefore

$$\theta^* = 1 - c, \quad x^* = 1 - c - \sigma\Phi^{-1}(c).$$

Takeaway: private noise selects one threshold equilibrium: collapse below θ^* , survival above it.

Appendix: empirical implication

The solved cutoff implies

$$A(\theta) = \Phi \left(\frac{x^* - \theta}{\sigma} \right).$$

Differentiating:

$$\frac{dA(\theta)}{d\theta} = -\frac{1}{\sigma} \phi \left(\frac{x^* - \theta}{\sigma} \right) < 0.$$

Validation target

Higher assigned regime strength should produce lower LLM joining.